

UE25CS251A
DIGITAL LOGIC AND COMPUTER DESIGN

TEAM DLCD

Department of CSE and CSE (AI & ML)

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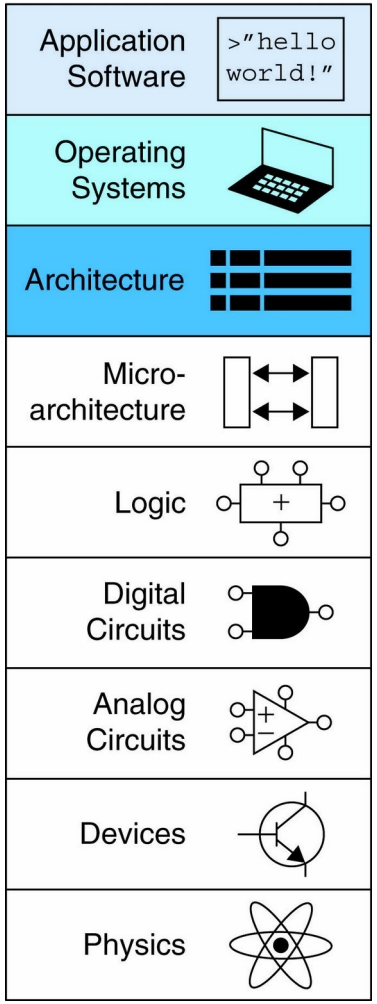
ARCHITECTURE-Introduction

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TOPICS

- Introduction
- Assembly Language
- Machine Language
- Programming
- Addressing modes
- Parallelism



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Introduction

What is Computer Architecture

Computer Architecture is the programmer's view of a computer. It defines how software communicates with hardware by specifying:

- Instruction Set Architecture (ISA)
- Operand locations (Registers and Memory)
- Data representation and instruction formats

Why Study Computer Architecture?

- Bridges the gap between hardware and software.
- Determines how programs are executed by the processor.
- Helps design efficient, compatible, and high-performance computer systems.
- Provides a foundation for understanding processor organization and programming.

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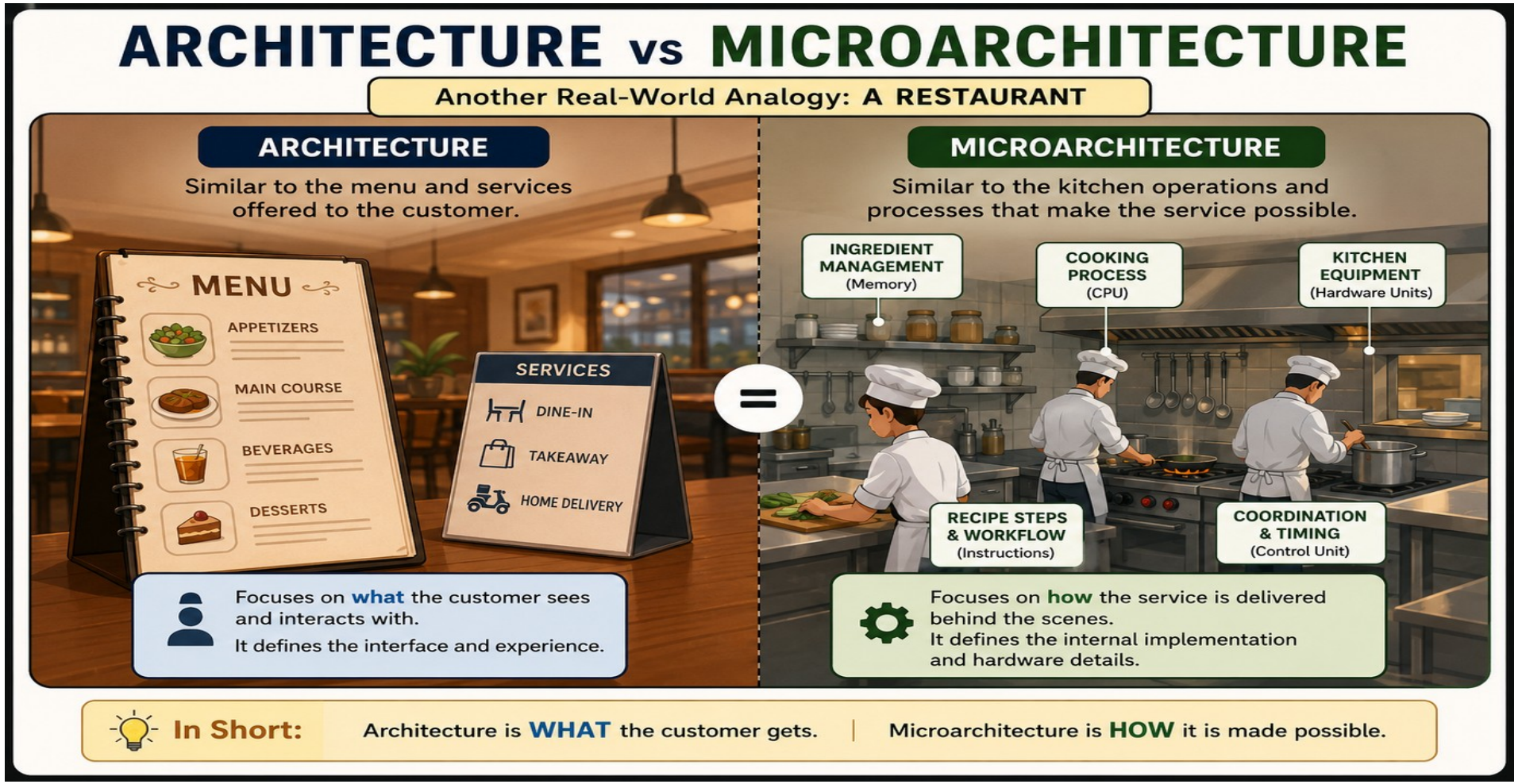
Architecture vs. Microarchitecture

Computer Architecture	Microarchitecture
Defines what the processor does	Defines how the processor is implemented
Visible to programmers	Internal hardware organization
Includes ISA, registers and memory	Includes ALU, datapath, control unit, pipeline, cache

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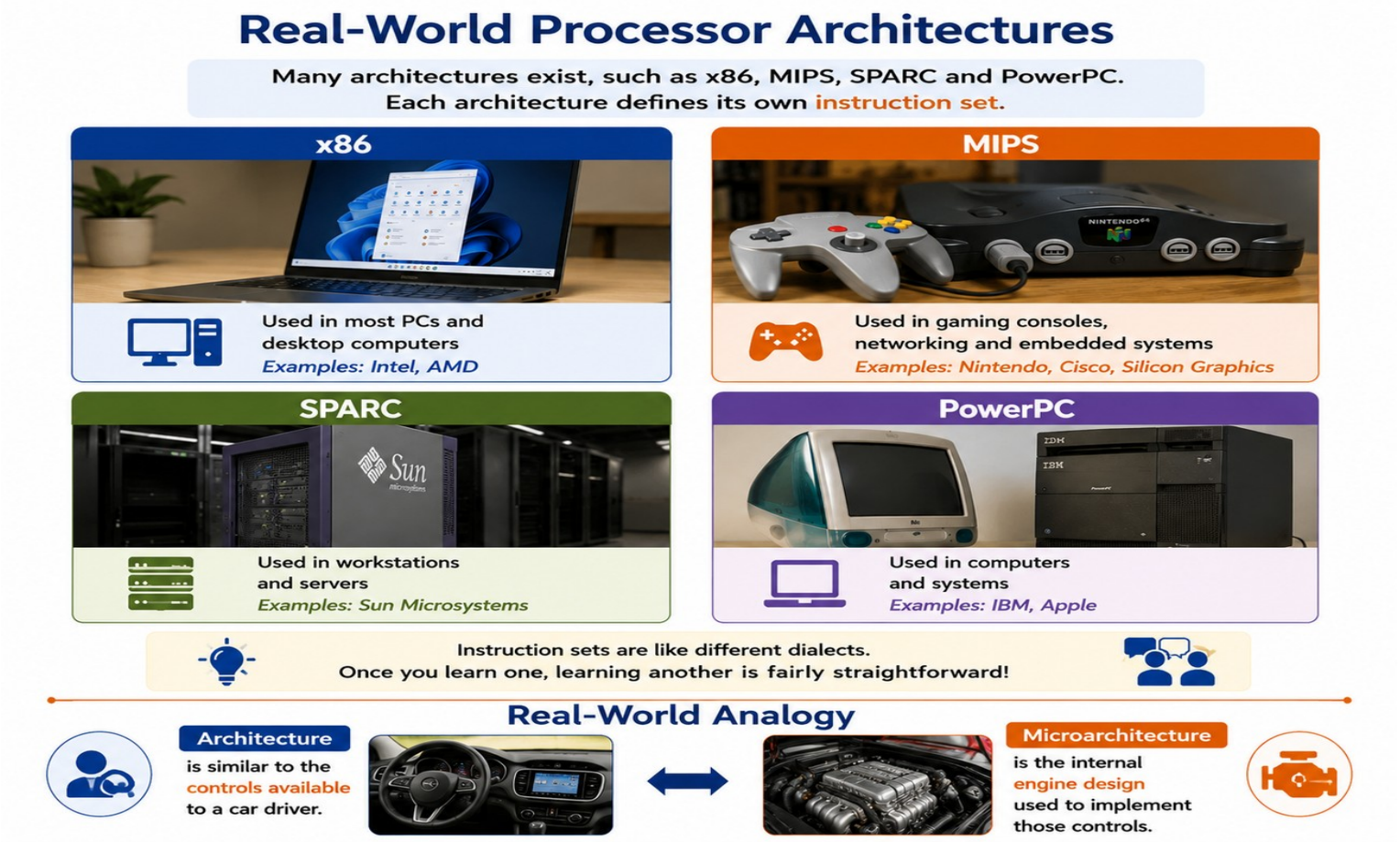
Introduction

- **Many Architectures Exist**: e.g. x86, MIPS, SPARC, and PowerPC
  - Each architecture defines its own instruction set
- **Instruction Sets Are Like Dialects**: once you learn one, learning another is fairly straightforward
- **Instructions**: the words of a computer's language; even complex applications compile down to simple instructions like add, subtract, and jump

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MCQ

- **Q1. What is “architecture” as defined in this chapter?**
  a. The programmer’s view of a computer, defined by the instruction set and operand locations
  b. The physical arrangement of transistors on a chip
  c. The way registers are wired to the ALU
  d. A hardware description language

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MCQ

- **Q2. Which of the following is the human-readable form of instructions?**
  a. Machine language
  b. Assembly language
  c. Binary code
  d. Microcode

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MCQ

- **Q3. Machine language instructions are ultimately encoded as:**
  a. Decimal numbers
  b. ASCII characters
  c. Binary (1's and 0's)
  d. Hexadecimal strings only

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MCQ

- **Q4. Who developed the MIPS architecture, and where?**
  a. John Hennessy and colleagues at Stanford
  b. Dennis Ritchie at Bell Labs
  c. Gordon Moore at Intel
  d. Alan Turing at Cambridge

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MCQ

- **Answer Key**

Q1: A Q2: B Q3: C Q4: A

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THANK YOU

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